

# Automation & Cypress Fundamentals

## Research & Practice

### Objective

By the end of this assignment, you should understand the core Cypress commands and be able to write simple automated tests using them.

For every concept below, answer the following:

1. What does it do?
  2. When would you use it?
  3. Write a simple example.
  4. Run the example in Cypress and attach a screenshot showing it works.
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## Part A: Test Structure

Research the following:

- `describe()`
- `it()`
- `before()`
- `beforeEach()`
- `after()`
- `afterEach()`

For each one explain:

- Purpose
  - Syntax
  - Example
  - A real-world use case
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## Part B: Assertions

Research:

- `expect()`
- `should()`

Explain:

- Difference between the two
  - When to use each
  - Three examples for each
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## Part C: Basic Cypress Commands

Find the Cypress command for each action.

| Action                 | Command | Example |
|------------------------|---------|---------|
| Visit a page           |         |         |
| Type into a textbox    |         |         |
| Click a button         |         |         |
| Clear a field          |         |         |
| Check a checkbox       |         |         |
| Uncheck a checkbox     |         |         |
| Select from a dropdown |         |         |
| Scroll to an element   |         |         |
| Upload a file          |         |         |
| Hover over an element  |         |         |
| Right click            |         |         |
| Double click           |         |         |

Press keyboard keys

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## Part D: Locators

Research how Cypress locates elements.

Explain and give examples of:

- `cy.get()`
- `cy.contains()`
- `.find()`
- `.children()`
- `.parent()`
- `.closest()`
- `.eq()`
- `.first()`
- `.last()`
- `.within()`

Then answer:

1. Why are IDs preferred?
  2. What are CSS selectors?
  3. What are data attributes (`data-cy`, `data-testid`)?
  4. Why are long CSS selectors discouraged?
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## Part E: Assertions Practice

Write examples using:

- `be.visible`
- `exist`
- `contain`
- `have.text`
- `have.value`
- `have.length`
- `be.enabled`
- `be.disabled`

- `be.checked`
  - `have.attr`
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## Part F: Working with Elements

Write Cypress examples for:

- Buttons
  - Text fields
  - Password fields
  - Checkboxes
  - Radio buttons
  - Dropdowns
  - Text areas
  - Links
  - Images
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## Part G: Waiting

Research the following:

- Automatic waiting
- Retry-ability
- `cy.wait()`
- Waiting for API requests
- Waiting for page loading

Answer:

1. Why is `cy.wait(5000)` considered bad practice?
  2. What is a better alternative?
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## Part H: Forms

Automate filling a form that contains:

- First Name
- Last Name
- Email
- Password
- Date Picker
- Dropdown
- Checkbox
- Radio Button
- Submit Button

Verify that the submission was successful.

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## Part I: Tables

Using any demo website:

Write tests to:

- Count rows
  - Count columns
  - Read data from a row
  - Click a button inside a row
  - Verify table contents
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## Part J: Browser Interactions

Research and write examples for:

- Browser Back
- Browser Forward
- Reload
- Open a new tab (how Cypress handles this)
- Browser alerts
- Confirmation dialogs
- Prompt dialogs

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## Part K: Keyboard Actions

Research examples for:

- Enter
  - Tab
  - Escape
  - Arrow Up
  - Arrow Down
  - Arrow Left
  - Arrow Right
  - Delete
  - Backspace
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## Part L: Scrolling

Write examples that:

- Scroll to the bottom of a page
  - Scroll to the top
  - Scroll to a specific element
  - Verify the element becomes visible
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## Part M: File Upload

Research:

- Why Cypress cannot upload files by default
  - The plugin or built-in approach commonly used
  - Upload a sample image or PDF
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# Part N: Mini Project

Choose one of the following demo websites:

- <https://practice.expandtesting.com/>
- <https://the-internet.herokuapp.com/>
- <https://demoqa.com/>
- <https://testautomationpractice.blogspot.com/>

Write Cypress tests that perform the following:

1. Visit the website.
  2. Fill out a form.
  3. Select values from a dropdown.
  4. Check and uncheck checkboxes.
  5. Select radio buttons.
  6. Upload a file.
  7. Scroll to a hidden element.
  8. Verify a success message.
  9. Use at least five different assertions.
  10. Organize your test using `describe()`, `it()`, and at least one hook (`beforeEach()` or `before()`).
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## Bonus Challenge (Optional)

Research and demonstrate:

- `cy.intercept()`
- `cy.fixture()`
- `cy.request()`
- Custom Cypress commands
- Page Object Model (POM)

Explain what each does and provide one working example.

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## Submission Requirements

Submit:

1. Your Cypress project (GitHub repository link).
2. A [README.md](#) explaining what you learned.
3. Screenshots or screen recordings showing successful test execution.
4. A short reflection (200–300 words) answering:
  - What was the easiest concept to understand?
  - What was the most challenging?